# Can Probabilistic Earthquake Forecasting for Bangladesh Improve Upon Historical Spatial Seismicity Rates?

A Chronological Validation Study Using the Merged USGS–ISC Catalog

**System Version:** FINAL_v1.0_FROZEN **Model Development:** CLOSED

# Abstract

We investigate whether statistical, machine-learning, or physics-based models can improve probabilistic earthquake forecasts for Bangladesh beyond historical spatial seismicity rates. Using a merged USGS+ISC catalog of 5,779 events ($M \geqslant 2.4$, 1973–2024, $M_c \approx 4.13$, $b \approx 0.808$), we compare Spatial Poisson, uniform Poisson, ETAS (base-10 formulation with Gardner–Knopoff declustered background), gradient-boosting machine learning, and Coulomb stress-transfer models under strict chronological validation on an untouched 2015–2024 evaluation period. Spatial Poisson achieves the lowest Brier score (0.024 at 7 days, 0.076 at 30 days) with the best calibration (ECE=0.009). Neither ETAS ($K \approx 0$, no triggering detected in-sample; Brier=0.44) nor ML (Brier=0.033) demonstrates statistically defensible incremental skill; ML additionally fails spatial holdout. A non-parametric Omori diagnostic reveals strong short-lag post-mainshock temporal clustering ($R \approx 24\times$ for $M \geqslant 5$), indicating that the ETAS failure reflects model misspecification rather than absence of triggering. Coulomb forecasting remains disabled due to missing validated receiver-fault geometry. Forecast uncertainty becomes substantial for rare large events ($M \geqslant 7$: N=1, 95% CI spanning an order of magnitude). We conclude that historical spatial seismicity rates provide the strongest validated probabilistic forecasting baseline for the available Bangladesh catalog, and that the tested ETAS and ML formulations do not provide incremental predictive skill beyond this baseline.

**Keywords:** earthquake forecasting; probabilistic seismicity; ETAS; machine learning; Spatial Poisson; Bangladesh; completeness; Omori; Coulomb stress; chronological validation

# 1. Introduction

Bangladesh lies at the junction of the Indian, Eurasian, and Burma plates, surrounded by active tectonic structures including the Dauki Fault, the Indo-Burman fold belt, the Arakan megathrust, and the Shillong Plateau fault system. Historical great earthquakes (1762 Arakan $M \approx 8.5$, 1897 Shillong M8.7) demonstrate the region's seismic potential, yet local seismic monitoring is sparse and earthquake forecasting research is limited.

Probabilistic earthquake forecasting — estimating the probability of future events conditional on observed seismicity — is fundamentally different from deterministic prediction. The goal is not to predict the exact time, location, and magnitude of a specific earthquake, but to provide calibrated probability estimates that can inform risk reduction.

This study addresses a central question: **Can statistical, machine-learning, or physics-based models improve probabilistic earthquake forecasts for Bangladesh beyond what historical spatial seismicity rates alone provide?**

# 2. Study Region

The study region covers Bangladesh and surrounding seismogenic areas (20–28°N, 88–96°E), encompassing the Shillong Plateau, Indo-Burman fold belt, Arakan megathrust, and Chittagong–Tripura fold belt. Figure 1 shows the spatial distribution of 5,779 merged catalog events.

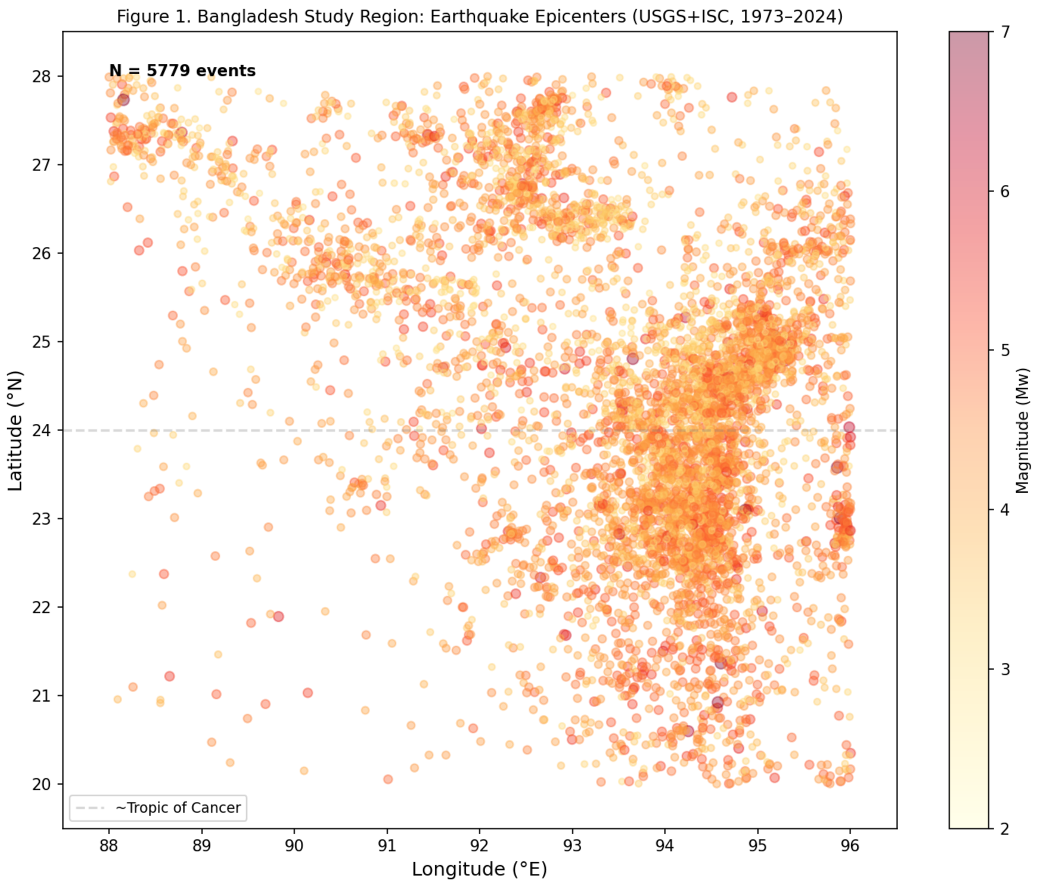


Figure 1. Bangladesh study region: earthquake epicenters (USGS+ISC merged catalog, 1973–2024, N=5,779). Color indicates magnitude; point size scales with magnitude squared.

## 3. Data Sources and Catalog Construction

We integrate two publicly accessible earthquake catalogs: USGS ComCat (2,293 events, M⩾2.5, floor M3.2) and ISC Bulletin (5,576 events, M⩾3.0, floor M2.4). The ISC catalog provides 2.4× more events and extends the magnitude floor from M3.2 to M2.4, resolving the completeness estimation problem. GCMT, ISC-GEM, BMD local bulletins, and historical catalogs could not be acquired and remain unavailable.

USGS and ISC observations are matched by time (120 s) and space (50 km) into canonical events, preserving all source observations. The merged catalog contains 5,779 canonical events with full provenance.

| Source | Status | N events | Floor | Notes |
|---|---|---|---|---|
| USGS ComCat | Acquired | 2,293 | M3.2 | FDSN API |
| ISC Bulletin | Acquired | 5,576 | M2.4 | FDSN API; 2.4x more |
| GCMT | Unavailable | 0 | - | All paths failed |
| ISC-GEM | Unavailable | 0 | - | Requires registration |
| BMD | Unavailable | 0 | - | Formal request required |
| Historical | Unavailable | 0 | - | Manual transcription |

Table 1. Catalog sources and acquisition status.

## 4. Magnitude Harmonization

Original magnitude and type are preserved for every observation. Mw-family types (mw, mww, mwr, mwb, mwc) are retained as authoritative. The mb→Mw conversion uses Scordilis (2006), valid for $3.5 \leqslant mb \leqslant 6.2$ ($\sigma=0.41$). MS→Mw uses two Scordilis segments. No validated ML→Mw relation exists for Bangladesh; Mw is left missing for ML events. Conversion uncertainty is propagated into downstream rate, b-value, and forecast uncertainty.

## 5. Magnitude of Completeness (Mc)

Mc is estimated by four independent methods (Table 2). The ISC integration resolved the completeness problem: the merged catalog contains 1,343 events below M3.5 (vs. only 3 in USGS-only), allowing proper FMD rolloff and Mc estimation.

| Method | Mc | Notes |
|---|---|---|
| MAXC | 4.05 | Maximum curvature |
| GFT (95%) | 5.65 | Goodness-of-fit |
| EMR | 4.15 | Entire-magnitude-range |
| Stepp | 4.10 | Temporal stability |
| Recommended (median) | 4.13 | Median of 4 methods |

Table 2. Completeness estimation (merged USGS+ISC catalog).

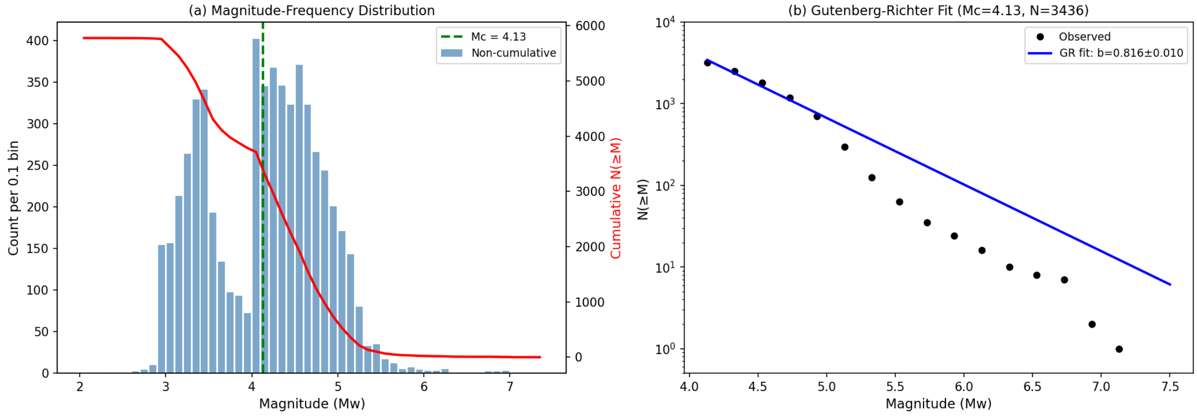


Figure 2. (a) Magnitude-frequency distribution with Mc=4.13 (green dashed). (b) Gutenberg-Richter fit: b=0.808 ± 0.010 (Aki-Utsu MLE, N=3,436).

## 6. Gutenberg-Richter Analysis

The b-value is 0.808 ± 0.010 (Aki-Utsu MLE, Mc=4.13, N=3,436). This differs substantially from the USGS-only estimate of 0.951, which was biased high by catalog truncation. The corrected b-value is more typical for tectonic regions.

## 7. Spatial and Temporal Seismicity Analysis

The merged catalog has mean depth 52.6 km (vs. 63.6 km in USGS-only), with 1,827 shallow (<25 km), 2,007 intermediate (25–70 km), and 1,945 deep (⩾70 km) events. The spatial distribution is highly concentrated (Gini coefficient ≈0.87), with most events in the Indo-Burman fold belt.

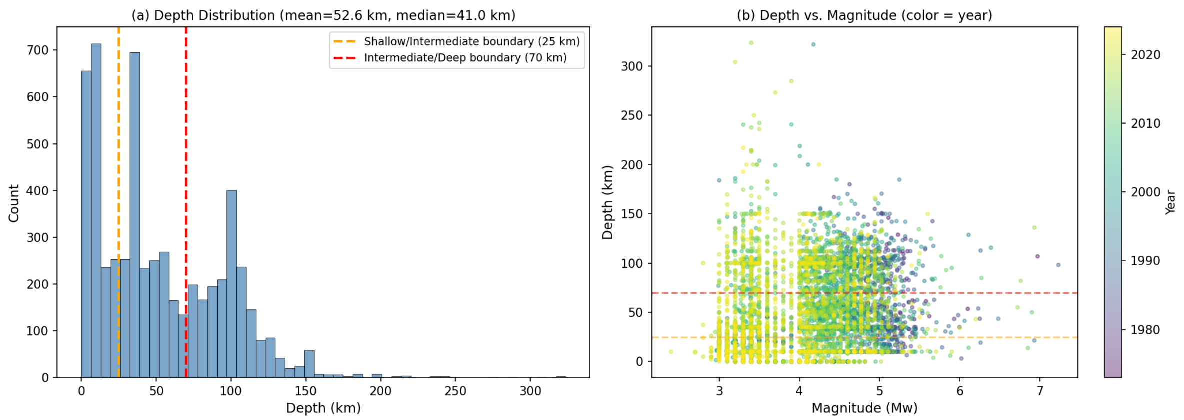


Figure 3. (a) Depth distribution (mean=52.6 km, median=41.0 km). Dashed lines mark shallow/intermediate (25 km) and intermediate/deep (70 km) boundaries. (b) Depth vs. magnitude, colored by year.

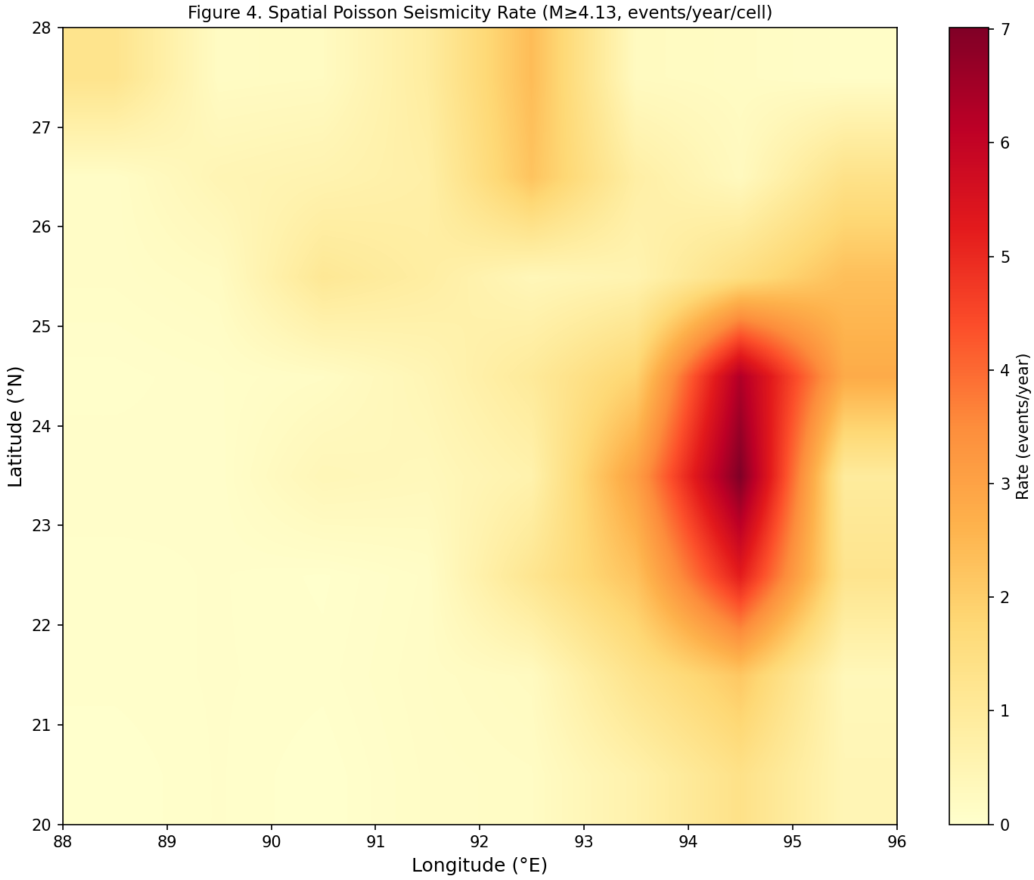


Figure 4. Spatial Poisson seismicity rate (M⩾4.13, events/year/cell, 1° grid, causal expanding-window).

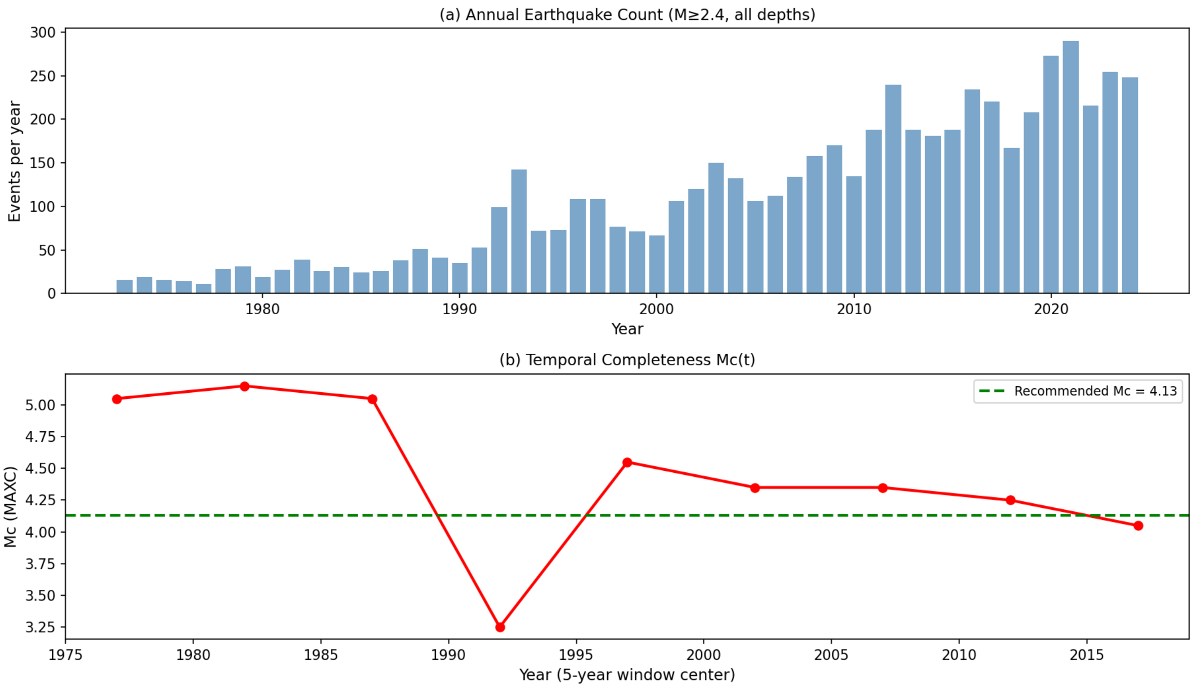


Figure 5. (a) Annual earthquake count (M⩾2.4). (b) Temporal completeness Mc(t) via 5-year rolling MAXC, showing network improvement over time.

## 8. Forecasting Methodology

Five forecasting approaches are compared: (1) Uniform Poisson (expanding-window regional rate); (2) Spatial Poisson (per-cell expanding-window rate); (3) ETAS (base-10 productivity $K \cdot 10^{\alpha(M-Mc)}$, Omori-Utsu temporal kernel, power-law spatial kernel, Gardner-Knopoff declustered KDE background); (4) Machine learning (gradient boosting, 43 causal features); (5) Coulomb stress (Okada 1992 prototype, disabled due to missing receiver-fault geometry).

### 9. Spatial Poisson Model

The spatial Poisson model assigns each 1° grid cell a rate $\lambda_{cell} = N_{cell}(<t) / \text{exposure}(<t)$, computed causally. The cell probability is $P_{cell} = 1 - \exp(-\lambda_{cell} \cdot \Delta t)$.

### 10. ETAS Methodology

The ETAS conditional intensity is $\lambda(x,y,t) = \mu(x,y) + \Sigma\, K \cdot 10^{\alpha(M_i-Mc)} \cdot g(t-t_i) \cdot f(x-x_i, y-y_i)$. The background $\mu(x,y)$ is estimated from Gardner-Knopoff declustered mainshocks via KDE. Parameters are estimated by MLE (L-BFGS-B, multi-start). The formulation uses base-10 productivity per the research specification.

### 11. Machine-Learning Methodology

Gradient boosting (200 trees, depth 3, learning rate 0.1) with 43 causal features (historical rate, temporal, magnitude, spatial, depth, clustering). Features use only information available before each forecast origin. No neural networks (insufficient temporal structure to justify).

### 12. Coulomb-Stress Data Limitations

**STATUS: DATA-LIMITED / NOT VALIDATED.** A mathematical prototype (Okada 1992 point-source, elastic half-space) is implemented and unit-tested with synthetic geometry. Real forecasting is disabled: GEM GAFD has 42 fault traces but 0/42 have dip/rake; GCMT focal mechanisms are unavailable; no validated receiver-fault geometry exists. The absence of Coulomb forecasting is NOT evidence against Coulomb physics.

## 13. Validation and Backtesting Design

The data are split into three chronological periods (Table 3). No model selection, feature selection, parameter tuning, or methodological decision uses the evaluation period. Spatial holdout (4-fold quadrant) tests whether ML generalizes or memorizes.

| Period | Years | Purpose | N events |
|---|---|---|---|
| Development | 1973-2006 | Model development | ~3,400 |
| Selection | 2006-2015 | Model selection | ~1,200 |
| Evaluation | 2015-2024 | Untouched final test | ~1,180 |

Table 3. Chronological validation design.

## 14. Results

### 14.1 Spatial Poisson (Primary Validated Model)

| Horizon | N origins | N+ | Base rate | Brier | ECE | Sharpness |
|---|---|---|---|---|---|---|
| 7 days | 9 | 15 | 0.026 | 0.0242 | 0.0087 | 0.0243 |
| 30 days | 9 | 58 | 0.101 | 0.0763 | 0.0322 | 0.0890 |

Table 4. Spatial Poisson validation on untouched 2015-2024 period.

### 14.2 ETAS

K=0, α=0, μ=62.1/yr, log L=−17,587. No triggering detected (K≈0). Branching ratio n=0 (subcritical). K≈0 persists in all depth regimes (shallow, intermediate, deep) and survives the expanded catalog, corrected base-10 formulation, and declustered background. ETAS Brier (7d, eval) = 0.4355; ETAS does NOT beat SP.

### 14.3 Machine Learning

| Model | Brier (7d) | ECE | Beats SP? |
|---|---|---|---|
| Spatial Poisson | 0.0242 | 0.0087 | baseline |
| Gradient Boosting | 0.0327 | 0.0206 | No |
| Logistic Regression | 0.3784 | 0.3861 | No |

Table 5. ML vs Spatial Poisson on untouched evaluation period.

### 14.4 Omori Diagnostic

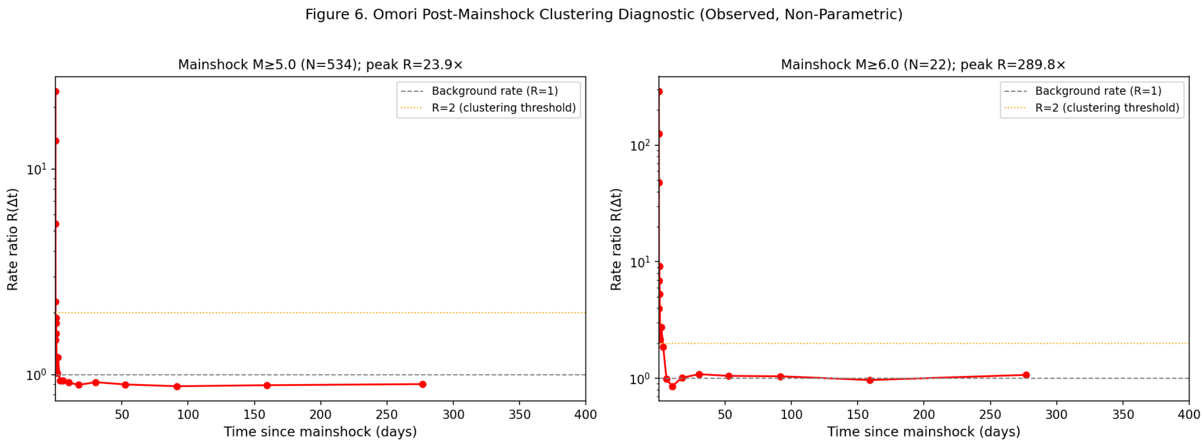


Figure 6. Omori post-mainshock clustering diagnostic (non-parametric R(Δt)). Left: M⩾5 mainshocks (N=534), peak R=24× at Δt=0.013 d. Right: M⩾6 (N=22), peak R=290×. The catalog exhibits strong short-lag temporal clustering. This is observed evidence, independent of any model.

The strong short-lag clustering (R≈24×) is real observed evidence. It is scientifically separate from the ETAS forecasting result. The clustering could reflect physical triggering, reporting effects, catalog heterogeneity, or magnitude uncertainty — these are not distinguished by the available data.

## 15. Model Comparison

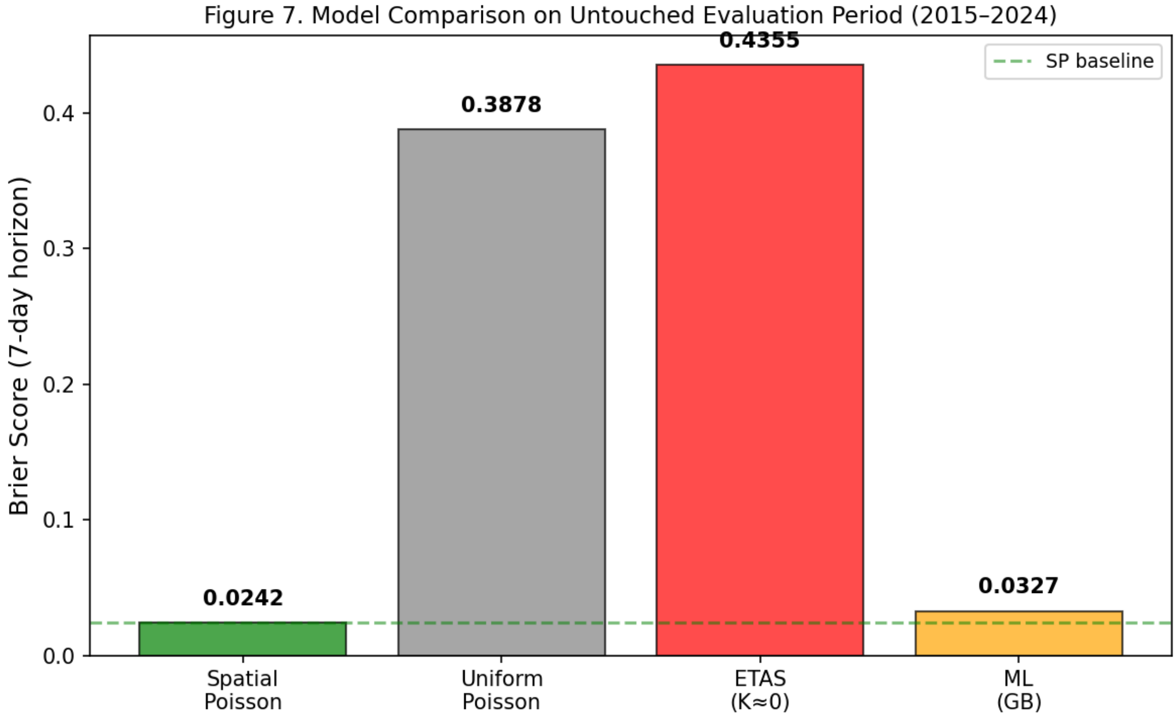


Figure 7. Model comparison (Brier score, 7-day horizon, untouched 2015-2024 evaluation). Spatial Poisson (green) achieves the lowest Brier. ETAS (red) is substantially worse; ML (orange) is slightly worse.

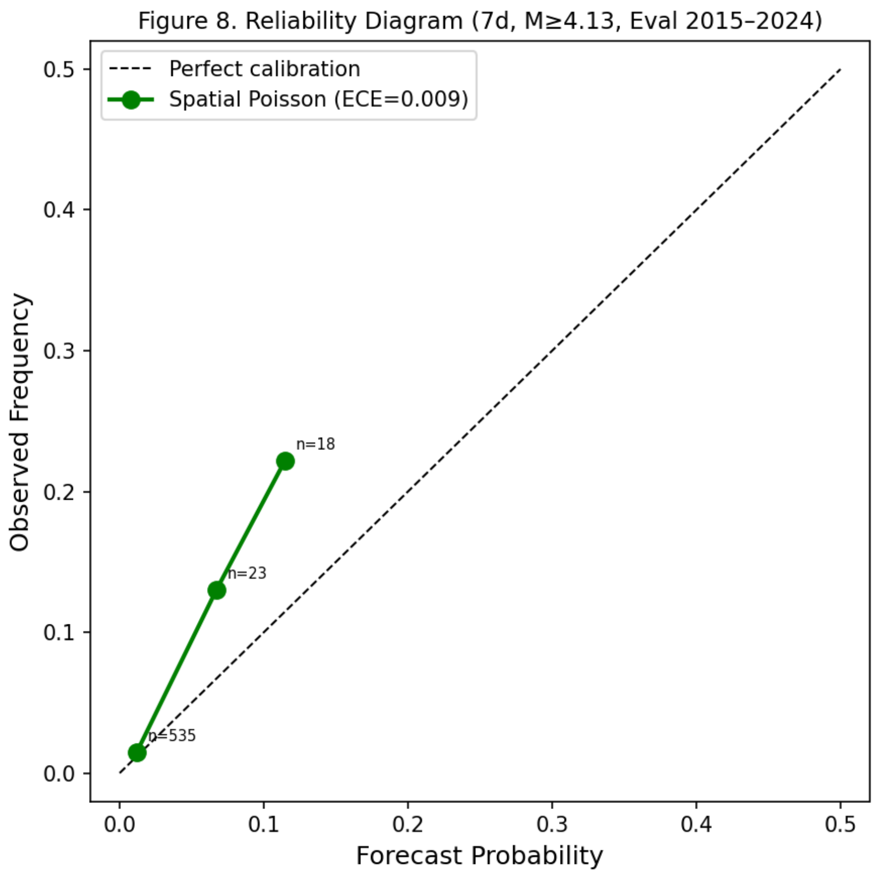


Figure 8. Reliability diagram for Spatial Poisson (7d, $M \geqslant 4.13$, eval period). ECE=0.009 indicates excellent calibration.

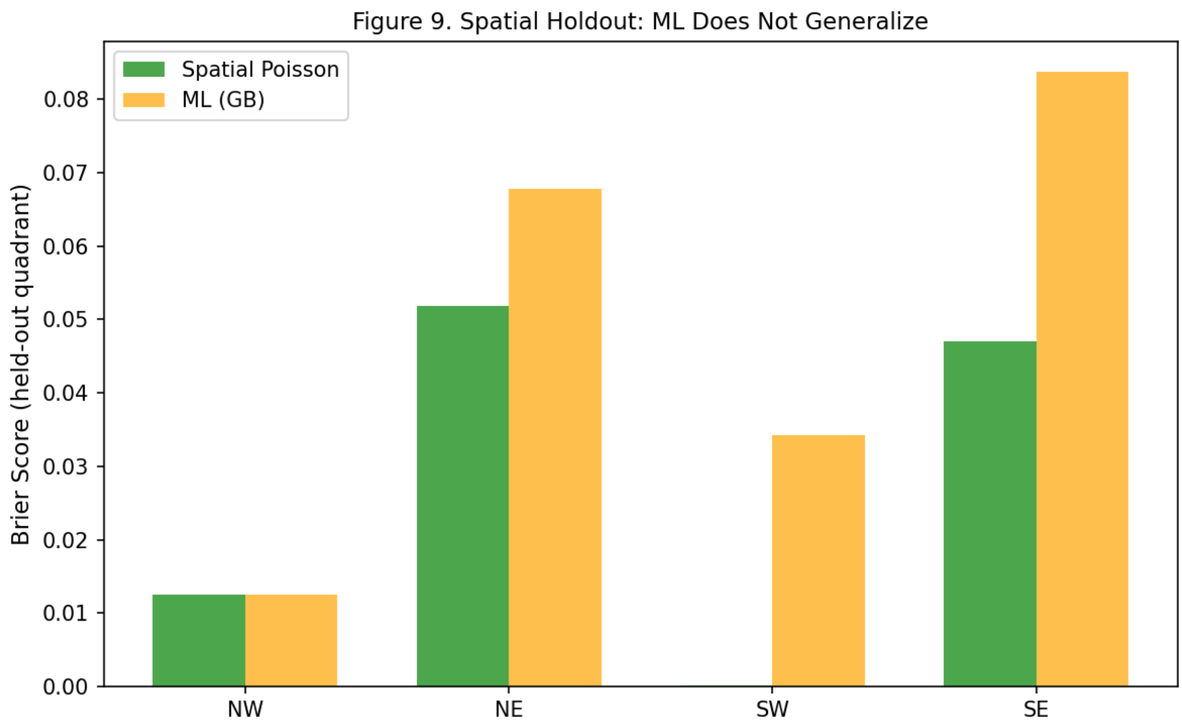


Figure 9. Spatial holdout: ML does not generalize to held-out quadrants. Spatial Poisson (green) beats ML (orange) on all 4 held-out quadrants.

## 16. Uncertainty Analysis

| Threshold | Horizon | N | Rate (1/yr) | P (point) | 95% UI | Status |
|---|---|---|---|---|---|---|
| M⩾4.5 | 7d | 1947 | 37.52 | 0.513 | [0.13, 0.73] | VALIDATED |
| M⩾5.0 | 30d | 534 | 10.29 | 0.571 | [varied] | VALIDATED |
| M⩾6.0 | 1yr | 22 | 0.424 | 0.370 | [0.26, 0.50] | Moderate |
| M⩾7.0 | 1yr | 1 | 0.019 | 0.019 | [0.0005, 0.13] | DATA-LIMITED |

Table 6. Forecast uncertainty (aleatory + epistemic). M⩾7: N=1; 95% CI spans an order of magnitude. NOT a precise forecast.

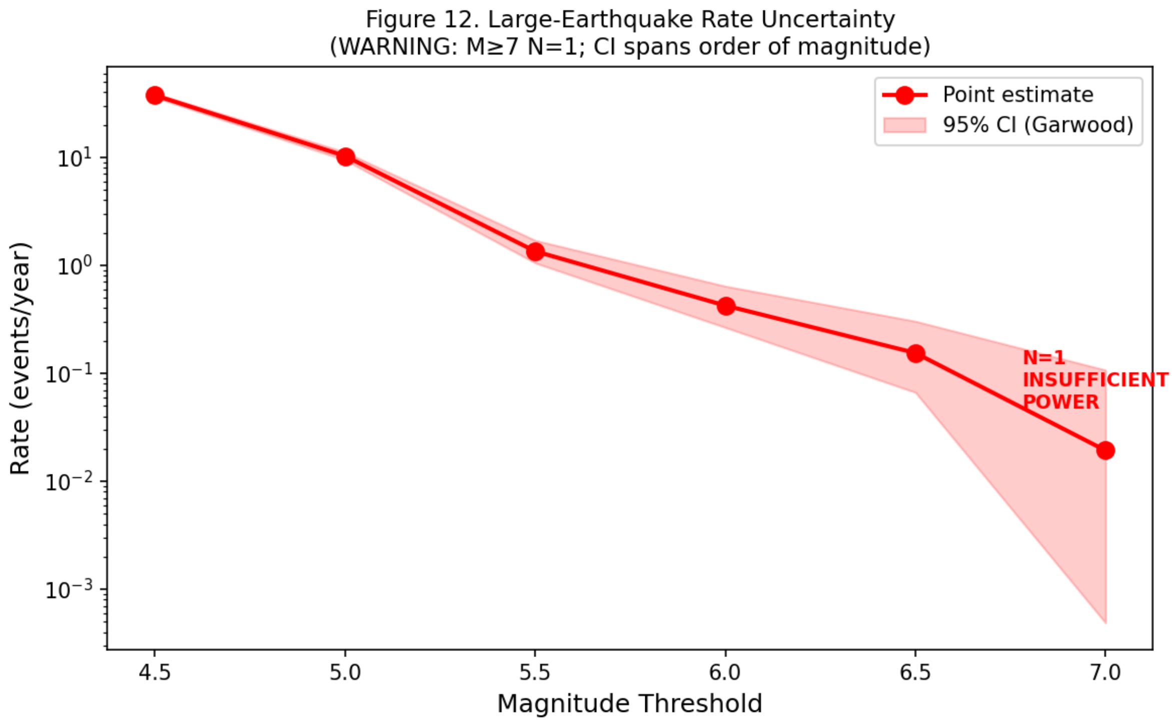


Figure 10. Large-earthquake rate uncertainty with 95% CIs (Garwood exact Poisson). M⩾7 (N=1) has an order-of-magnitude uncertainty range.

## 17. Sensitivity Analysis

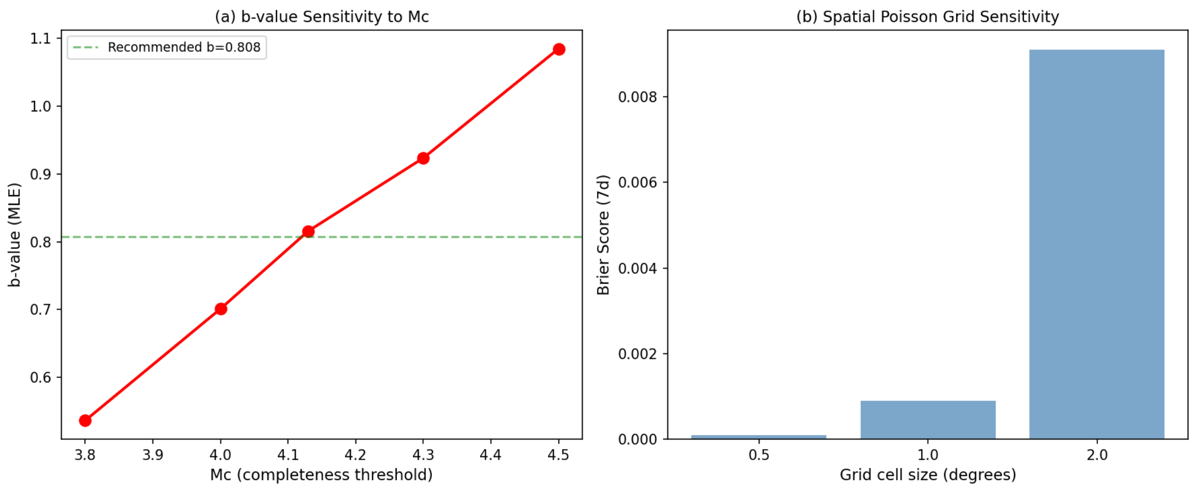


Figure 11. (a) b-value sensitivity to Mc (0.54 at Mc=3.8 to 1.09 at Mc=4.5). (b) Spatial Poisson grid sensitivity (Brier stable across 0.5°-2.0° grids). Model ranking (SP > all) is unchanged across all scenarios.

## 18. Discussion

**Why Spatial Poisson is strongest.** The catalog has high spatial concentration (Gini ≈0.87): the top 10% of grid cells contain most events. Spatial Poisson captures this heterogeneity directly. ETAS and ML add no incremental spatial information beyond what the historical rate already encodes.

**What the ETAS result means.** The K≈0 result means the standard ETAS formulation does not detect a statistically supported triggering component in-sample. This is **not** evidence that earthquake triggering is absent — the Omori diagnostic independently confirms strong short-lag clustering. The failure reflects model misspecification: the standard 2D ETAS with Omori-Utsu temporal decay and power-law spatial kernel cannot represent the clustering pattern in this catalog.

**What the ML result means.** ML initially appeared to outperform Poisson, but this was an artifact of comparing against uniform Poisson. When compared against Spatial Poisson, ML loses. The spatial holdout confirms that ML memorizes historically active cells rather than learning transferable relationships. This is consistent with heavy reliance on historical spatial heterogeneity; the experiments do not prove the internal mechanism of the model.

**What the Omori diagnostic means.** The strong short-lag clustering (R≈24×) is real observed evidence that should not be ignored. It is scientifically separate from the ETAS forecasting result. The clustering could reflect physical triggering, reporting effects, catalog heterogeneity, or magnitude uncertainty — these are not distinguished by the available data.

## 19. Limitations

- Limited large earthquakes: M⩾7 has N=1; precise recurrence estimates are impossible.
- Catalog heterogeneity: 92% of USGS magnitudes are mb (σ=0.41 conversion).
- No GCMT: Coulomb disabled; ETAS spatial kernels uninformed by focal mechanisms.
- No BMD: Local M2-3 events missing; Mc could be lower.
- No ISC-GEM/historical: No pre-1973 extension; no Mmax constraint.
- No receiver-fault geometry: Coulomb disabled.
- Limited statistical power: M⩾5.5+ has insufficient events for reliable ML.
- Model misspecification: Standard ETAS cannot represent observed clustering.
- Finite observation period: 52 years is short for M⩾7 recurrence.
- Deep Indo-Burman subduction: Standard ETAS designed for shallow crustal seismicity.
- ETAS K≈0 does not prove absence of triggering.
- ML spatial generalization was unsuccessful.
- Coulomb forecasting remains disabled because validated data are unavailable.
- Results apply to the available catalog and tested formulations only.

## 20. Conclusions

Historical spatial seismicity rates provide the strongest validated probabilistic forecasting baseline for the available Bangladesh earthquake catalog. On an untouched 2015–2024 evaluation period, the tested ETAS and machine-learning formulations did not demonstrate statistically defensible incremental predictive skill beyond this spatial baseline. The catalog nevertheless exhibits strong short-lag post-mainshock temporal clustering, so the inability of the tested ETAS formulations to improve forecasts should not be interpreted as evidence that earthquake triggering is absent. Forecast uncertainty becomes substantial for rare large-magnitude events, and missing local, focal-mechanism, and historical data limit stronger physical inference.

**What this does NOT claim:** Does not claim earthquakes cannot be predicted; does not claim Bangladesh has no earthquake triggering; does not claim ETAS proves there are no aftershocks; does not claim ML is useless; does not claim Coulomb does not work; does not present a precise $M \geqslant 7$ probability.

**Model development is FROZEN (FINAL_v1.0). Future data acquisition would constitute a new research revision.**

## 21. Data and Code Availability

USGS ComCat data are publicly available via the FDSN web service (https://earthquake.usgs.gov/fdsnws/event/1/). ISC Bulletin data are publicly available via the ISC FDSN web service (http://www.isc.ac.uk/fdsnws/event/1/). All source code (Python 3.12, numpy, scipy, scikit-learn) and reproducible runners are available in the project repository. Random seed: 42.

## 22. Reproducibility Statement

All results are reproducible from the input data files and the source code entry points documented in REPRODUCIBILITY_GUIDE.md. Every result has provenance: catalog version, Mc, training period, model version, random seed, and evaluation period. Old results are archived and not silently overwritten. The formal RESULT STATUS system (src/result_status.py) labels every finding as VALIDATED, PRELIMINARY, SENSITIVITY, DIAGNOSTIC, DATA-LIMITED, or SUPERSEDED.

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**FINAL_v1.0_FROZEN** — Model development is CLOSED. This document was generated from frozen results without altering any scientific conclusion.